

CMPE 485 – Introduction to Game Programming

Term Project – Phase I Submission

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Option A (Single Challenge) March 23, 2026

1 Chosen Technical Challenge

GPU Instancing & Draw Call Optimization in Unity

The selected challenge is the performance impact of draw calls on CPU/GPU utilization when rendering large numbers of identical objects, and how *GPU Instancing* mitigates this bottleneck. A draw call is issued by the CPU to the GPU for every rendered object; when thousands share the same mesh and material, the redundant per object CPU overhead becomes the primary frame rate limiter, a well known bottleneck in open world and crowd simulation scenarios. GPU Instancing batches all such objects into a single draw call, dramatically reducing CPU overhead and improving throughput. The final report will include a rationale supported by case analyses from game development forums, technical blogs, and academic papers to underline the significance of this optimization in the industry.

2 Experiment Design Overview

A controlled Unity scene will be created containing a flat terrain populated with a variable number of identical static meshes (e.g. a medium-poly tree or rock). Two rendering modes will be toggled via a runtime UI switch:

- **Baseline:** Standard rendering — no batching, no instancing.
- **Optimized:** GPU Instancing enabled on the shared material.

Independent variable: Object count $N \in \{100, 500, 1000, 5000, 10000\}$.

Dependent variables: FPS, CPU frame time (ms), GPU frame time (ms), draw call count.

Control: Fixed camera position, constant lighting, no physics, consistent hardware environment (Apple M4 Pro, unified memory). Each configuration will be recorded over a 30-second window using the Unity Profiler and a custom in-game FPS counter.

3 Project Plan

Milestone	Deadline	Tasks
Phase I	23 Mar 2026	Topic selection, roadmap document (this submission).
Scene Setup	30 Mar 2026	Build test scene; implement runtime object spawner; add instancing toggle; verify draw call counts in Frame Debugger.
Profiling & Data	14 Apr 2026	Run all $N \times 2$ configurations; collect Unity Profiler snapshots; export CSV data (FPS, CPU/GPU frame times, draw call counts).
Analysis & Writing	03 May 2026	Produce figures/tables; analyse results; write full technical report.
Final Submission	10 May 2026	Submit report, Unity project source code, and short demonstration video.
Presentation	11–13 May 2026	Prepare slides; in-class presentation during lecture hours.